

Optical Character Recognition (OCR) of Low-Resource Languages

Lightweight OCR for North Sámi Heritage Digitization

Jan Paweł Musioł

Supervised by Aparna S. Varde & Juan Heredia

The Maersk Mc-Kinney Møller Institute
University of Southern Denmark

Bachelor Thesis 2026 • IISA 2026 full paper • SCAI 2026 abstract

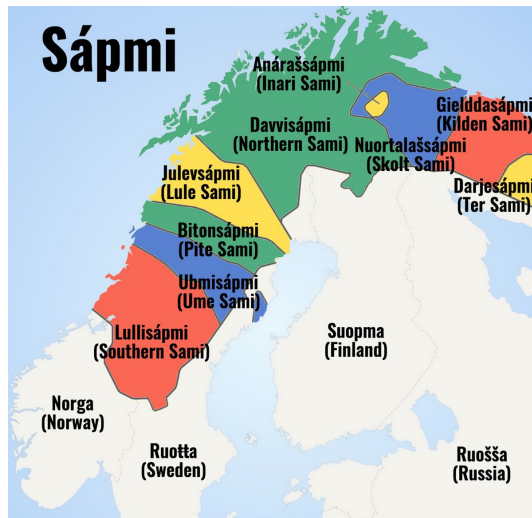
June 2026

Outline

- 1 Motivation & Research Question
- 2 Background
- 3 Methodology & What was built
- 4 Results
- 5 Reflection & Conclusion

What are Sámi languages & Why this matters

- Nine mutually unintelligible Sámi languages across Northern Europe
- North Sámi has ~25,000 speakers
- classified by UNESCO as **definitely endangered**
- Heritage preservation is critical for culture survival → digitization/OCR makes it possible
- Technology should be available for everyone



What is OCR?

OCR: Optical Character Recognition

- Task of converting scanned documents into machine readable text
- Essential for digitization and preservation of culture
- Archives, libraries

mijáv juohkkaláhkáj kreatijvvan.



mijáv juohkkaláhkáj kreatijvvan.

The gap

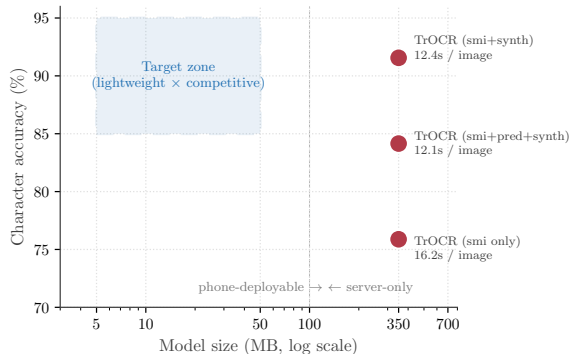
*Comparative study: Tesseract vs Transcribus vs TrOCR.*¹

State of the art:

Transformer-based OCR (TrOCR)

- 91.45% character accuracy
- 350 MB on disk
- 12+ s / image on CPU

Not well fitted for
resource-constrained deployments.



Accuracy vs. size/latency.

¹Enstad et al. (2025). *Comparative analysis of optical character recognition methods for Sámi texts from the National Library of Norway*. Proc. NoDaLiDa/Baltic-HLT, pp. 98–108.

Research question & contributions

Research question

Can a lightweight OCR architecture approach Transformer-level accuracy for North Sámi while enabling practical deployment on modest hardware?

What makes North Sámi hard for OCR

- 1 Lack of publicly available real scanned documents**
- 2 Seven special diacritics**
á, č, đ, ŋ, š, ƭ, ž
- 3 Agglutinative morphology**
Hundreds of inflected forms per verb,
large type-token ratio → model cannot
memorise word shapes.
- 4 Consonant gradation**
giella (“language”) vs. *giela* (“of
language”) — a single dropped “l”
changes the meaning.

Northern Sámi Special Characters

Á/á <i>áhku (grandmother)</i>	Č/č <i>čáhci (water)</i>	Đ/đ <i>odda (new)</i>	Ŋ/ŋ <i>eaggals (english)</i>
Š/š <i>šaidi (bridge)</i>	Ʀ/ƞ <i>mátolaš (possible)</i>	Ž/ž <i>ieža (other)</i>	

Two architectural approaches for OCR

CRNN² based OCR: CNN

(Convolutional Neural Network) + RNN
(Recurrent Neural Network) + CTC
(Connectionist Temporal Classification)

- Reads image from left to right
- CNN: extracts column features
- RNN: sequence encoder
- CTC: head + Loss, align length-varying input to output

Transformer based OCR (TrOCR)³:

ViT (Vision Transformer) encoder +
autoregressive decoder

- Image is split into patches and processed with self-attention
- Token-by-token generation based on the encoded image
- Pre-trained on massive image corpora

²Shi, Bai, Yao (2017). *An end-to-end trainable neural network for image-based sequence recognition and its application to scene text recognition*. IEEE Trans. PAMI, 39(11), pp. 2298–2304.

³Li et al. (2023). *TrOCR: Transformer-based optical character recognition with pre-trained models*. Proc. AAAI, 37, pp. 13094–13102.

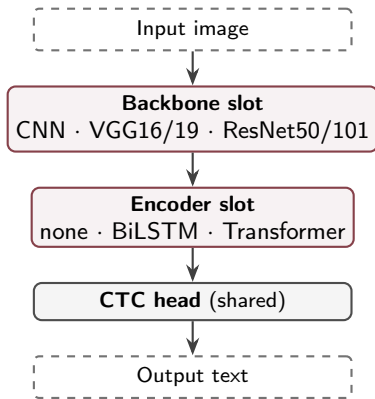
A modular OCR pipeline

Based on CRNN: 5 backbones $\times$ 3 encoders, shared CTC head.

Three swappable slots:

- Backbones: Feature extraction
- Encoders: Sequence Context
- Decoder: Shared CTC head

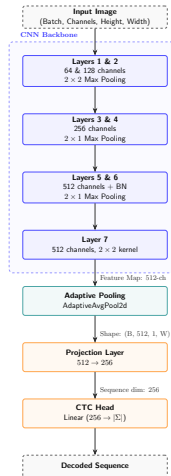
Due to compute budget **only four** variants were trained. Three failed to converge.



The headline model: CNN-CTC (ctc_simple)

7-layer CNN, trained from scratch

- 1** Preprocessing: Greyscale, normalization, resize
- 2** Feature extraction: 2x2 max pooling + 2x1 asymmetric pooling
- 3** No encoder: extracted features passed directly to CTC head (adaptive pooling + projection layer)
- 4** Sequence decoding: 395 classes, greedy decoding, collapses repeated characters and removes blank tokens.



Experimental setup

Train Språkbanken synthetic Sámi OCR dataset (National Library of Norway), ~307k line images (276k training + 31k hyperparameter tuning), 32×800px.

Evaluation 1,048 samples

- 1,000 in-domain synthetic lines (Språkbanken validation split)
- 48 out-of-domain parallel Northern Sámi–English sentences from Turi's 1910 "An Account of Sámi"

Metrics CER (Character Error Rate), WER (Word Error Rate), exact match, normalised accuracy, Character/Word Accuracy (100 – CER/WER)

Main benchmark results

Model	Char. Acc. % ↑	Word Acc. % ↑	sec / img	Size (MB)
trocr_smi_synth	91.45	71.84	12.41	~350
ctc_simple	87.76	62.65	0.033	23
trocr_smi_pred_synth	84.15	54.24	12.08	~350
trocr_smi	75.88	36.50	16.23	~350

Char. Acc. = 100 - CER Word Acc. = 100 - WER

Results for ctc_simple: 87.76% char. acc. (within 3.69 pp of best TrOCR), 380× faster, 15× smaller.

Throughput: 109,000 img/hour for ctc_simple vs 290 img/hour for TrOCR

Real-life: Scanning 1 milion of pages takes <10 hours on CPU vs. ~144 days for TrOCR.

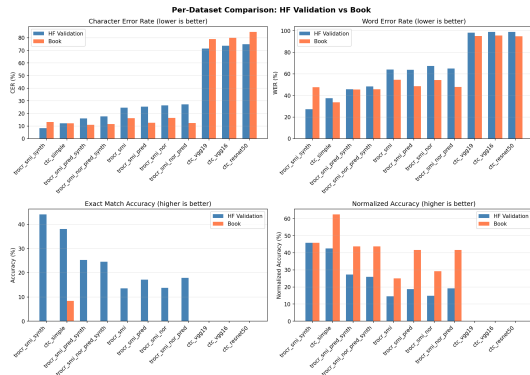
In-domain vs Out-of-domain evaluation

On 1910s historical text (Friis orthography):

- TrOCR degrades 4.84 pp
- ctc_simple accuracy stays flat
- On normalised accuracy ctc_simple performs better

But!

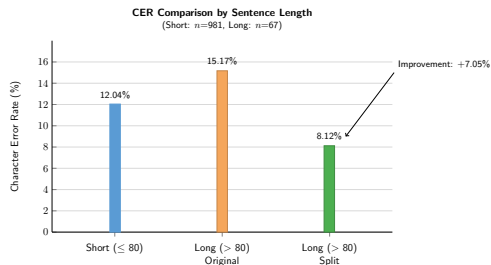
$n = 48$ samples. Not conclusive, might suggest...
The CNN-CTC is more stable but results lack statistical power.



Conjunction-based sentence splitting

- Sentences with over 80 characters have higher CER.
- Mitigation: Split at North Sámi conjunctions (*ja*, *muhto*, *dahje*, *go*, *ahte*, *jos*), OCR each segment, rejoin.
- Long-sentence CER drop: -7.05 pp

Proof of concept: operates on ground-truth, the cut images are re-rendered without noise.



End-to-end OCR → MT pipeline

- CNN-CTC output is feed to TartuNLP API (sme to eng)
- Translation accuracy is moderate
- Character errors land on parts carrying gramatical role
- Useful for indexing content discovery but not full translation

Metric	Score
BLEU ↑	10.40
chrF ↑	33.85
TER ↓	76.30%

BLEU (Bilingual Evaluation Understudy) chrF (character n-gram F-score) TER (Translation Edit Rate)

Example (CER 1.2%):

GT (sámi): *ja áldoeallu lea váralaš dan galggalii sáhttit nu siivvut reainnidit go lea mátolaš*

Pred (sámi): *... sáhttit nu siivut reainnidit ...* [one “v” dropped]

GT (en): “and it is a dangerous time for the cow herd, they must be herded as carefully as possible.”

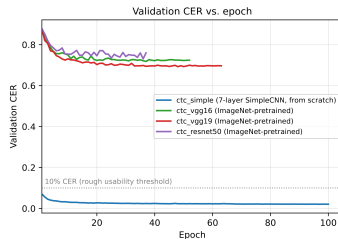
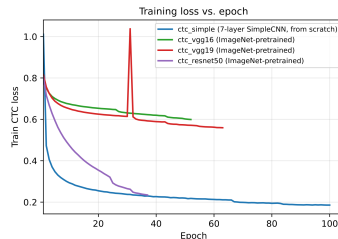
Pred (en): “and the livestock is dangerous, it should be able to be removed as safely as possible.”

Pretrained backbones failed training

Three ImageNet pretrained backbones failed to converge.

- VGG16/VGG19: *underfitting* - the training loss barely moved
- ResNet50: *overfitting* - the training loss went down but validation loss didn't move

Hypothesis: ImageNet's weights were pre-trained to detect object-level features and not character-level.



Conclusion

Were the goals achieved?

- Lightweight CNN-CTC OCR model for North Sami, achieving 87.76% character accuracy, within 3.69 pp CER of state of the art, 380× faster and 15 × smaller with stable historical-text behaviour
- Makes the deployment easier for communities, which the technology should serve.

Applications & future work

Applications

- Mobile / edge OCR for community digitalization
- Large (milions) scanned documents indexing
- Joint OCR + reconstruction of occluded text (Figure)

Future work

- Real scanned corpus
- Giellatekno morphological post-correction
- Layout analysis for full pages
- Transfer to other Sámi varieties (Inari, Skolt, Lule)



*Damaged-text reconstruction proof of concept:
OCR + diffusion inpainting on an artificially occluded
line.*

Thank you

Questions?

`jamus23@student.sdu.dk`
`github.com/magwrap/north-sami-ocr`
`huggingface.co/magwrap/cnn-ctc-ocr-sme`